

Real time object detection using open CV

What is object detection?

Object detection is a computer technology related to computer vision and image processing that deals with detecting instances of semantic objects of a certain class in digital images and videos.

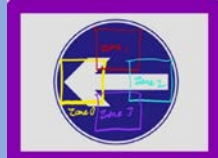
Goal

Detect Signs Real Time Applications

- Robotics
- Self driving cars

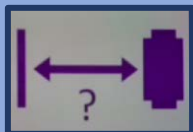
Method

- Template matching



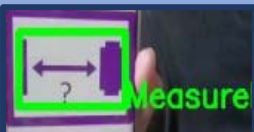
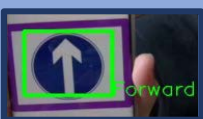
Template

0	0	0
0	1	0
0	1	0



Real time detection

0	1	1	0	0
1	1	0	0	0
0	0	0	1	0
0	1	0	1	0
1	0	1	1	1



Face Recognition

Applications

- Security
- Surveillance

Method

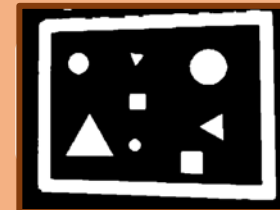
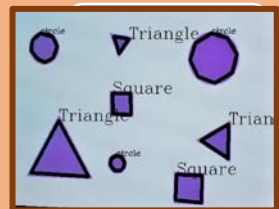
- Haar Cascade



Shape Counting

Method

- Contours



Results

- Thresholds and parameters requires trial and error to gain accuracy.
- Delay was significant enough to effect real time object detection in the moving car.
- Limited by capability of hardware.

Conclusion

OpenCV is a versatile library for image processing that can also function on weak systems - like the raspberry pi - with plenty of pre-made algorithms that make more advanced image detection accessible to new learners.